# `llama-modchip`: Hardware-Aware Mathematical and Kernel Co-Design for 30B–284B Sparse MoE Inference at 256K Context on Commodity CPUs and GPU+CPU Hybrids

**Paolo C**
Independent Systems & Machine Learning Researcher
`https://github.com/synchopate/llama-modchip`

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## Abstract

Running frontier Sparse Mixture-of-Experts (MoE) language models at a 262,144-token (256K) context window is widely assumed to require multi-GPU clusters with high-bandwidth HBM memory. On commodity CPUs (50–90 GB/s dual-channel DDR5), standard inference engines fail due to four compounding bottlenecks: (1) quantizing 256-expert SwiGLU layers below 2 bits/weight without calibration data collapses reasoning, while standard 2-bit formats exceed 64 GiB RAM on 284B-parameter models; (2) neural draft models (0.5B–1B) for speculative decoding read 2–3 GB of weights per draft cycle, saturating the DDR5 bus and flushing the CPU's victim L3 cache; (3) four-level page-table walks across 4–14 million 4 KB pages thrash the L2 STLB; and (4) naive linear rescaling during dynamic MoE expert pruning distorts the variance ratio against dense shared experts.

We present `llama-modchip` (`v2.4.0`), an open-source hardware-aware inference engine that co-designs quantization formats, SIMD kernels, cache/TLB hierarchy management, draftless chess-engine speculative search, and norm-preserving low-rank/MoE mathematics. First, we introduce **Block-64 2-Level Hierarchical Non-Linear Quantization (`Fractal-64`, 3.75 bpw)** with single-cycle AVX-512 `vprorvq` and AVX2 `vpsllvd+vpmaddwd` sub-scale fusion, paired with **Analytical Second-Order `SwiGLU-H2` Hessian Compression**, shrinking the 284.33B-parameter `DeepSeek-V4-Flash` (13.17B active) from 76.87 GiB to **56.56 GiB** on disk and **58.9 GiB total `VmRSS` at full 256K context** on a single 64 GB CPU. Second, we replace neural draft models with **`RouteCrack + Stockfish-PVS v2`**, a 1.64 MB L2/L3-resident speculative engine combining an 8-level FNV-64 hash cascade with 4-way Zobrist transposition tables and Principal Variation Search, achieving **96.39% live draft acceptance** (48.39 tok/s average / 60.40 tok/s peak on a 16-core Zen 5 CPU, and **111.62 tok/s** on a 64-core Threadripper PRO 7985WX) on `Nemotron-3.5-30B-A3B`. Third, we prove that because routed MoE expert vectors in $\mathbb{R}^{4096}$ are nearly mutually orthogonal, tail expert pruning must use **$L_2$-quadrature rescaling** rather than linear $L_1$ rescaling, cutting next-token KL-divergence by **58.1%** (to 0.0042 nats / +0.42% PPL at Top-5) and pushing pure single-token CPU decode on `DeepSeek-V4-Flash 284B` from 5.91 tok/s to **9.06 tok/s (lossless)**, **9.68 tok/s (Top-5 $L_2$)**, and **10.12 tok/s (Top-4 $L_2$)**.

## 1 Introduction

Modern frontier language models increasingly adopt sparse Mixture-of-Experts (MoE) architectures paired with state-space or latent-attention compression to decouple total parameter capacity from per-token active compute. Two prominent examples illustrate this design space:

1. `NVIDIA-Nemotron-3.5-Lightning-30B-A3B` [1]: A 52-layer hybrid architecture (30.2B total, 3.0B active parameters) comprising 23 Mamba-2 State-Space Model (SSM) layers, 23 Sparse MoE layers (128 routed experts with top-6 routing plus 1 shared expert), and only 6 Grouped-Query Attention (GQA) layers (2 KV heads per layer, 12 KV heads total).

2. `DeepSeek-V4-Flash-0731` [2]: A 43-layer frontier MoE Transformer (284.33B total, 13.17B active parameters) combining 64-head Multi-Head Latent Attention (MLA, compressed KV latent rank 512, RoPE dimension 64, query LoRA rank 1024, and 8 output head-groups of rank 1024), sparse indexer compressors (`CSA`/`LID`/`HCA`), 4-stream Sinkhorn-20 Hyper-Connections (hc = 4), and 256 routed experts (top-6 active plus 1 shared expert per layer), where layers 0..2 employ deterministic token-ID hash routing (`tid2eid`) and layers 3..42 employ learned $\sqrt{\text{softplus}}$ top-$k$ routing.

In principle, activating only 3.0B to 13.17B parameters per token makes these architectures attractive for local deployment on single-socket workstations without data-center GPUs. Prior systems for consumer hardware,

such as PowerInfer [6] (which exploits ReLU activation sparsity via offline neuron profiling), KTransformers [7] (which offloads MoE experts to CPU while running attention on a GPU), SqueezeLLM [8] (sensitivity-based non-uniform quantization requiring calibration gradients), and FlashAttention-2 [9] (GPU-centric IO-aware attention tiling), have advanced local inference significantly. However, deploying modern SwiGLU/Mamba-2/MLA MoE models at a full 262,144-token (256K) context window on a pure 64 GB RAM CPU box (or a consumer 8–16 GB GPU + 16–64 GB RAM hybrid) without offline calibration datasets exposes four compounding bottlenecks that neither baseline `llama.cpp` [3] nor prior hybrid engines address:

- **The 64 GiB Capacity & Non-Divisibility Wall**: Even at aggressive 2-bit quantization (`IQ2_XXS`), `DeepSeek-V4-Flash` requires 76.87 GiB on disk, triggering immediate Linux OOM termination on a 64 GB host. Meanwhile, on `Nemotron-3.5-30B-A3B`, expert dimensions ($1856 \times 2688$) are not divisible by standard K-quant super-block sizes ($QK_K = 256$), forcing `llama-quantize` to fall back to 4.5 bpw `Q4_0` (17.60 GiB).
- **The CPU Speculative Bandwidth Paradox**: On HBM3 GPUs (3.35 TB/s), a 1B neural draft model executes 5 steps in $< 1$ ms. On dual-channel DDR5 CPUs ($\sim$ 72 GB/s), streaming 600 MB of 1B draft weights 5 times reads 3.0 GB from DRAM ($\sim$ 40 ms)—slower than executing the 30B MoE target model itself (1.8 GB active), while evicting hot recurrent state and KV blocks from the 64 MB L3 victim cache.
- **Microarchitectural TLB & Graph-Rebuild Stalls**: Mapping 18–57 GiB of weights in default 4 KB pages creates 4.5–14.8 million page table entries (PTEs), exceeding the 3,072-entry L2 STLB of AMD Zen 5 by $1,500\times$–$4,800\times$. Furthermore, dynamic compressor topology changes in `DeepSeek-V4-Flash` trigger full 4,551-node compute-graph rebuilds on 7 out of every 8 decode tokens.
- **Unexamined Linear & Orthogonal Redundancy**: Nearly 31% of per-token DRAM traffic in `DeepSeek-V4-Flash` (991.6 MB/tok) is spent in the unactivated linear chain $\mathbf{W}_{o,b}\mathbf{W}_{o,a}$, while naive linear weight renormalization during dynamic MoE tail pruning distorts the amplitude ratio between routed experts and the dense shared expert.

### 1.1 Contributions

This paper makes four tightly coupled contributions, all implemented in open-source C/C++ and verified across x86-64 (AMD Zen 5, 8-CCD Zen 4 Threadripper, Intel AVX2), ARM64 (AWS Graviton3, Apple M4 Pro), and GPU+CPU hybrids (Linux/Windows CUDA & Vulkan):

1. **`Fractal-64` Quantization, `SwiGLU-H2` Analytical Hessian Compression, & AVX-512/AVX2 Kernel Co-Design (§2)**: We design a 30-byte Block-64 hierarchical non-linear format (3.75 bpw) whose 4-bit sub-scales fuse at zero instruction overhead into AVX2 `vpmaddwd` and AVX-512 `vprorvq` (`_mm512_rorv_epi64`), and derive a closed-form second-order SwiGLU Hessian metric that compresses `DeepSeek-V4-Flash 284B` to 56.56 GiB (58.9 GiB `VmRSS` at 256K context).
2. **Mathematical Co-Design for `DeepSeek-V4-Flash`: SVD Spectrum of $\mathbf{W}_{o,b}\mathbf{W}_{o,a}$ & $L_2$-Quadrature MoE Tail Pruning (§3)**: We prove that because routed expert outputs in $\mathbb{R}^{4096}$ are nearly orthogonal, pruned router weights must be rescaled in quadrature ($\alpha_{L_2}$) rather than linearly ($\alpha_{L_1}$), reducing next-token KL-divergence by 58.1% (to 0.0042 nats at Top-5) while preserving hash-routed lexical layers ($il = 0..2$). We also characterize the exact singular value spectrum of $\mathbf{W}_{o,b}\mathbf{W}_{o,a}$ (99.82% cosine similarity at rank $r = 768/1024$).
3. **Draftless Chess-Engine Speculative Decoding (`RouteCrack + Stockfish-PVS v2`, §4)**: We formulate the CPU-MoE speculative wall-clock equation and introduce a 1.64 MB L2/L3-resident drafter using 4-way Zobrist transposition tables and Principal Variation Search, unlocking Singular Horizon Extensions up to $M = 7$ with 98.74% marginal precision at $D = 5..7$.
4. **Cross-Platform Memory Hierarchy & Hybrid GPU+CPU Orchestration (§5)**: We combine 2 MB Transparent HugePages, bidirectional `_MM_HINT_NTA` victim-L3 isolation, 3D L1D cache tiling, zero-rebuild dynamic `CSA/LID` node skipping, 8-slot `Q8` activation caching, deterministic 4C/CCD multi-chiplet pinning, and live-VRAM Windows/Linux tensor-role splitting (`-n-cpu-moe`).

## 2 Quantization & SIMD Co-Design

### 2.1 Block-64 `Fractal-64` (`Q2_0`, 3.75 bpw)

In `Nemotron-3.5-30B-A3B`, the MoE projections have inner dimensions $1856 = 29 \times 64$ and $2688 = 42 \times 64$. Because neither is divisible by $QK_K = 256$, standard K-quants (`Q3_K`, `IQ3_S`) cannot quantize the expert tensors. We introduce `Fractal-64` (mapped to the 64-weight `GGML_TYPE_Q2_0` slot), which encodes 64 weights in exactly **30 bytes (3.75 bits/weight)**:

- `ggml_half d` (2 bytes): FP16 super-block scale.
- `uint8_t scales[4]` (4 bytes): Eight 4-bit unsigned sub-scales $m_s \in \{0..15\}$ (representing shifted multipliers $\tilde{m}_s = m_s + 4 \in \{4..19\}$) covering eight 8-weight micro-blocks.
- `uint8_t qs[16]` (16 bytes) and `uint8_t qh[8]` (8 bytes): Lower 2 bits and upper 1 bit of 64 3-bit indices $q_i \in \{0..7\}$, mapped onto a symmetric non-linear codebook $\mathcal{C} = \{-14, -8, -4, -1, +1, +4, +8, +14\}$ fitted to the empirical Kurtosis of SwiGLU expert weights.

**Zero-Overhead SIMD Sub-Scale Fusion.** A naive implementation would convert each 8-weight dot product to FP32 before multiplying by $\tilde{m}_s$. Instead, against `Q8_0` activations $x_i \in [-127, 127]$, the maximum magnitude of an 8-element inner product is $8 \times 14 \times 127 = 14,224$. However, in our AVX2/AVX-512 kernel (`ggml_vec_dot_q2_0_q8_0_generic`), we accumulate pairs of products via `_mm256_maddubs_epi16` (`vpmaddubsw`), whose 16-bit signed lanes have a worst-case bound of $2 \times 14 \times 127 = 3,556$. Multiplying by the maximum sub-scale $\tilde{m}_s \le 19$ yields $3,556 \times 19 = 67,564$, whereas `vpmaddwd` (`_mm256_madd_epi16(p01, vmult01)`) widens directly to 32-bit signed integers (`INT32_MAX` $= 2.14 \times 10^9$) while summing adjacent pairs! Thus, replicating $\tilde{m}_s$ into 16-bit lanes and feeding it as the second operand of `vpmaddwd` fuses sub-block rescaling and horizontal pairwise reduction in a **single clock cycle with zero extra instructions**. On Zen 5, extracting the 64-bit upper bit-plane `qh[8]` is further accelerated via single-instruction AVX-512 variable 64-bit rotation `_mm512_rorv_epi64` (`vprorvq`), reducing 10M-block contraction time from 86.5 ms to **55.8 ms (+55% speedup)**.

## 2.2 Analytical Second-Order `SwiGLU-H2` Compression for 284B MoE

To fit `DeepSeek-V4-Flash` (284.33B parameters) inside $< 60$ GiB of RAM without a multi-terabyte calibration run, we observe that each routed expert computes $\mathbf{y}^{(e)}_{\text{FFN}}(\mathbf{x}) = \mathbf{W}^{(e)}_{\text{down}}\mathbf{h}^{(e)}(\mathbf{x})$, where the intermediate SwiGLU activation is the gated bilinear product:

$$\mathbf{h}^{(e)}(\mathbf{x}) = \text{SiLU}\Big(\mathbf{W}^{(e)}_{\text{gate}}\mathbf{x}\Big) \odot \Big(\mathbf{W}^{(e)}_{\text{up}}\mathbf{x}\Big). \tag{1}$$

When quantizing the down-projection $\mathbf{W}^{(e)}_{\text{down}} \to \mathbf{W}^{(e)}_{\text{down}} + \Delta\mathbf{W}^{(e)}_{\text{down}}$, the expected output reconstruction error $\mathbb{E}\|\Delta\mathbf{y}^{(e)}_{\text{FFN}}\|_2^2 \approx \sum_{j=1}^{d_{\text{ff}}} \|\Delta\mathbf{W}^{(e)}_{\text{down}}[:, j]\|_2^2 \cdot \mathbb{E}[(h^{(e)}_j(\mathbf{x}))^2]$ is weighted by the second moment of each intermediate channel $j$. Under isotropic unit-variance pre-activations $\mathbb{E}[\mathbf{x}\mathbf{x}^\top] \propto \mathbf{I}$, this yields a closed-form data-free Gauss-Newton / diagonal Hessian importance weight for expert $e$, channel $j$:

$$I_{e,j} = \mathbb{E}\Big[(h^{(e)}_j(\mathbf{x}))^2\Big] \propto \Big\|\mathbf{W}^{(e)}_{\text{gate}}[j,:]\Big\|_2^2 \cdot \Big\|\mathbf{W}^{(e)}_{\text{up}}[j,:]\Big\|_2^2. \tag{2}$$

In `scripts/transcode_dsv4.c`, we dequantize the paired Gate and Up expert slices simultaneously, pass $I_{e,j}$ as the importance vector to `quantize_row_iq1_s` when quantizing `ffn_down_exps` on bulk routed layers, retain high-sensitivity anchor layers ($\{0, 1, 26, 28, 41, 42\}$ in `IQ2_XXS`; $\{2, 4, 5, 8, 27, 30, 32, 40\}$ in `IQ1_M`), and transcode all dense `Q8_0` MLA and Indexer projections (`attn_q_b`, `attn_output_a`, `attn_output_b`, `attn_compressor_*`, `indexer_compressor_*`) to `Fractal-64` (`Q2_0`, 3.75 bpw). This shrinks the 3-shard GGUF from 76.87 GiB down to **56.56 GiB** ($-26.4\%$) and enables full 256K-context execution in **58.9 GiB `VmRSS`** on a 64 GB CPU host.

# 3 Mathematical Structure & Low-Rank / MoE Co-Design

## 3.1 Spectral Decay of the Unactivated Linear Chain $\mathbf{W}_{o,b}\mathbf{W}_{o,a}$

During single-token decode of `DeepSeek-V4-Flash`, $\sim$ 3.20 GB of weights are streamed from DDR5 per token. Profiling reveals that **31.0% of all DRAM traffic (991.6 MB/tok, or 23.06 MB/layer across 43 layers)** occurs in the MLA output projection (`deepseek4.cpp`), which applies two consecutive `Q2_0` matrices across $G = 8$ head-groups ($r_o = 1024$ inner dimension per group, $d = 4096$ model dimension) with **zero non-linearity, zero normalization, and zero bias** between them:

$$\mathbf{y}_{\text{attn}} = \sum_{g=1}^{8} \mathbf{W}^{(g)}_{o,b}\Big(\mathbf{W}^{(g)}_{o,a}\mathbf{x}_g\Big), \tag{3}$$

where $\mathbf{x}_g \in \mathbb{R}^{4096}$ is the concatenated output of the 8 MLA heads ($8 \times 512 = 4096$) in group $g$, $\mathbf{W}^{(g)}_{o,a} \in \mathbb{R}^{1024\times4096}$, and $\mathbf{W}^{(g)}_{o,b} \in \mathbb{R}^{4096\times1024}$.

De-quantizing the actual GGUF tensors (`blk.{0,3,4,10}.attn_output_{a,b}.weight`) and computing the exact Singular Value Decomposition $\mathbf{M}_g = \mathbf{W}^{(g)}_{o,b}\mathbf{W}^{(g)}_{o,a} = \mathbf{U}_g\mathbf{\Sigma}_g\mathbf{V}_g^\top$ across all 8 head-groups demonstrates rapid spectral decay ($\sigma_0 = 15.398 \to \sigma_{768} = 0.901 \to \sigma_{1023} = 0.048$; in early layers, the bottom 180 singular values fall below $10^{-3}$). As shown in Table 2, rank-768 truncation ($-25.0\%$ DRAM $= -247.9$ MB/tok, saving $\sim 6.9$ ms/tok) retains **99.64% of Frobenius energy** and achieves **0.998204 output cosine similarity**.

Table 1: Exact operator-by-operator single-token decode profile (`LLAMA_PROFILE_OPS`, $N = 1$, 4,551 graph nodes) and 512-token prompt prefill (`pp512`) on `DeepSeek-V4-Flash-0731-Fractal` (284.33B total / 13.17B active, 3 transcoded GGUF shards = 56.56 GiB from the 4-shard 76.87 GiB `IQ2_XXS` source) on a single 16C/32T AMD Ryzen 9 9955HX (Zen 5, 0 GPUs).

| Operator / Subsystem (43 Layers) | Types | Calls/Tok | v2.3.0 Init | Pre-Micro-Opt | v2.4.0 Lossless | v2.4.0 Top-4 L2 |
|---|---|---|---|---|---|---|
| 1. MLA & Shared Expert `MUL_MAT` (`wq_b`, `wo_a/b`, `shexp`) | `q2_0` | 512 | 78.40 ms | 62.20 ms | **61.81 ms** | **61.81 ms** |
| 2. Routed MoE `MUL_MAT_ID` (Bulk `SwiGLU-H2`) | `iq1_s` | 87 | 25.10 ms | 22.79 ms | **22.27 ms** | **15.17 ms** (−33.4%) |
| 3. Routed MoE `MUL_MAT_ID` (Anchor Layers) | `iq2_xxs` | 18 | 4.50 ms | 4.04 ms | **4.01 ms** | **2.75 ms** (−31.9%) |
| 4. Routed MoE `MUL_MAT_ID` (Mid-Sensitivity Layers) | `iq1_m` | 24 | 2.10 ms | 1.81 ms | **1.80 ms** | **1.16 ms** (−35.9%) |
| 5. Output Head (`lm_head`) + Router Gates | `q4_K/q6_K/f32` | 298 | 11.20 ms | 9.29 ms | **9.05 ms** | **9.05 ms** |
| 6. `HADAMARD` (FWHT), `RMS_NORM`, `HC_PRE/POST`, `FA` | `f32/f16` | 1,810 | 19.60 ms | 10.12 ms | **9.85 ms** | **8.90 ms** |
| 7. Graph Rebuild & Dynamic `CSA/LID` Compressor Skip | Graph/C++ | 1 | 28.40 ms | 11.67 ms | **1.54 ms** | **0.00 ms** |
| **Total Single-Token Decode Latency** ($N = 1$) | **All** | **4,551** | **169.30 ms** | **121.92 ms** | **110.33 ms** | **98.84 ms** |
| **Pure Single-Token Decode Throughput** (`tok/s`) | — | — | 5.91 tok/s | 8.20 tok/s | **9.06–9.95 tok/s** (+53.3%–68.4%) | **10.12–10.74 tok/s** (+71.2%–81.7%) |
| **Batched Prompt Prefill Throughput** (`pp512`) | — | 5,375 | 24.70 tok/s | 41.58 tok/s | **43.16–44.36 tok/s** (+77.1%) | **43.16–44.36 tok/s** (+77.1%) |

Table 2: Exact SVD truncation analysis of the linear operator $\mathbf{M}_g = \mathbf{W}_{o,b}^{(g)}\mathbf{W}_{o,a}^{(g)}$ across 8 head-groups in `DeepSeek-V4-Flash`.

| Rank $r/1024$ | DRAM Saved | Frobenius Energy | Cosine Sim | Rel. $L_2$ Err |
|---|---|---|---|---|
| $r = 1024$ (Full) | 0 MB/tok | 100.00% | 1.000000 | 0.00% |
| $r = 768$ (−25.0%) | −247.9 MB/tok | **99.64%** | **0.998204** | 5.99% |
| $r = 640$ (−37.5%) | −371.9 MB/tok | 98.61% | **0.992999** | 11.80% |
| $r = 512$ (−50.0%) | −495.8 MB/tok | 96.02% | 0.979718 | 20.02% |

## 3.2 Orthogonal Geometry of MoE Tail Pruning & $L_2$-Quadrature Rescaling

In each layer $l \in \{0..42\}$ of `DeepSeek-V4-Flash`, the FFN block sums one dense shared expert $\mathbf{e}_{\text{shared}}(\mathbf{x})$ and $K = 6$ routed experts $\mathbf{e}_{k_1}(\mathbf{x}), \ldots, \mathbf{e}_{k_6}(\mathbf{x}) \in \mathbb{R}^d$ ($d = 4096$):

$$\mathbf{y}_{\text{FFN}}(\mathbf{x}) = \mathbf{e}_{\text{shared}}(\mathbf{x}) + \sum_{i=1}^{K} w_i\, \mathbf{e}_{k_i}(\mathbf{x}), \quad \sum_{i=1}^{K} w_i = 1.5, \tag{4}$$

where $w_1 \geq w_2 \geq \cdots \geq w_K > 0$ are the sorted router weights. Logging $w_{1..6}$ across all 43 layers reveals a fundamental structural dichotomy:

1. **Hash-Routed Lexical Layers ($l \in \{0, 1, 2\}$)**: Selected via 6 deterministic token-ID hash tables (`ffn_gate_tid2eid`). Their weights are nearly flat (e.g., $l = 2$: $[0.270, 0.261, 0.258, 0.257, 0.255, 0.199]$). Pruning any expert in $l \in \{0, 1, 2\}$ discards 1/6th of the token's lexical representation and corrupts numeric/hexadecimal constants. Thus, $l \in \{0, 1, 2\}$ must remain unpruned ($k = 6$).

2. **Learned-Router Layers ($l \in \{3..42\}$)**: Selected via $\sqrt{\text{softplus}}$ top-$k$ routing. The weights obey a steep power law (e.g., $l = 19$: $[0.762, 0.202, 0.156, 0.145, 0.119, 0.117]$; $l = 8$: $[0.446, 0.386, 0.367, 0.120, 0.092, 0.088]$). Across $l \in \{3..42\}$, the top-4 experts hold 81.2% of $\sum w_i$, the top-5 hold 91.0%, and the 6th expert carries only 6%–9% ($< 4\%$ of total FFN energy including $\mathbf{e}_{\text{shared}}$).

**Proposition 1** (Suboptimality of Linear $L_1$ Rescaling vs. $L_2$-Quadrature Rescaling)**.** *Let $\mathbf{e}_{k_1}(\mathbf{x}), \ldots, \mathbf{e}_{k_K}(\mathbf{x}) \in \mathbb{R}^d$ ($d = 4096$) be zero-mean expert outputs with approximately equal second moments $\mathbb{E}\|\mathbf{e}_{k_i}\|_2^2 = \sigma_e^2$ and near-mutual orthogonality $\mathbb{E}\langle \mathbf{e}_{k_i}, \mathbf{e}_{k_j}\rangle \approx 0$ for $i \neq j$, and let $\mathbf{e}_{shared}(\mathbf{x})$ be orthogonal to the routed experts with $\mathbb{E}\|\mathbf{e}_{shared}\|_2^2 = \sigma_s^2$. When truncating the routed sum to $1 \leq k \leq K$ experts with a rescaling factor $\alpha > 0$, $\hat{\mathbf{y}}_{FFN}(\alpha) = \mathbf{e}_{shared} + \alpha \sum_{i=1}^{k} w_i \mathbf{e}_{k_i}$:*

1. *The minimum mean-squared error ($L_2$-optimal) projection onto span$(\mathbf{e}_{k_1}, \ldots, \mathbf{e}_{k_k})$ is achieved at $\alpha_{proj} = 1$, whereas preserving the exact expected squared norm $\mathbb{E}\|\hat{\mathbf{y}}_{FFN}(\alpha)\|_2^2 = \mathbb{E}\|\mathbf{y}_{FFN}\|_2^2 = \sigma_s^2 + \sigma_e^2 \sum_{i=1}^{K} w_i^2$ entering the downstream `RMSNorm` requires $L_2$-**quadrature rescaling**:*

$$\alpha_{L_2} = \sqrt{\frac{\sum_{i=1}^{K} w_i^2}{\sum_{i=1}^{k} w_i^2}}. \tag{5}$$

2. *Conversely, conventional linear $L_1$-sum rescaling $\alpha_{L_1} = \frac{\sum_{i=1}^{K} w_i}{\sum_{i=1}^{k} w_i}$ strictly inflates the variance of the routed MoE branch relative to the unscaled shared expert $\mathbf{e}_{shared}$ for all $1 \leq k < K$ by the factor:*

$$\Gamma(w) = \frac{\alpha_{L_1}^2}{\alpha_{L_2}^2} = \frac{\left(\sum_{i=1}^{K} w_i\right)^2}{\left(\sum_{i=1}^{k} w_i\right)^2} \cdot \frac{\sum_{i=1}^{k} w_i^2}{\sum_{i=1}^{K} w_i^2} > 1. \tag{6}$$

*Proof.* By mutual orthogonality, $\mathbb{E}\|\mathbf{y}_{\text{FFN}} - \hat{\mathbf{y}}_{\text{FFN}}(\alpha)\|_2^2 = \sigma_e^2 \left[(1-\alpha)^2 \sum_{i=1}^{k} w_i^2 + \sum_{i=k+1}^{K} w_i^2\right]$, which is minimized at $\alpha = 1$, while setting $\mathbb{E}\|\sum_{i=1}^{k}(\alpha w_i)\mathbf{e}_{k_i}\|_2^2 = \sigma_e^2 \alpha^2 \sum_{i=1}^{k} w_i^2 = \sigma_e^2 \sum_{i=1}^{K} w_i^2$ yields $\alpha = \alpha_{L_2}$. For typical learned-router weights at $k = 4$ ($\sum_{i=1}^{6} w_i = 1.50$, $\sum_{i=1}^{4} w_i = 1.22$, $\sum_{i=1}^{6} w_i^2 = 0.4806$, $\sum_{i=1}^{4} w_i^2 = 0.4414$), linear rescaling gives $\alpha_{L_1} = 1.2295$ (+23.0% amplitude inflation, $\Gamma = 1.388$), whereas quadrature rescaling gives $\alpha_{L_2} = \sqrt{0.4806/0.4414} = 1.0435$ (+4.35%). □

**Remark 1** (Positive Directional Correlation $\rho$ and Downstream `RMSNorm` SNR)**.** *If co-selected experts share a small positive pairwise directional correlation $\rho =$*

$\mathbb{E}\langle \mathbf{e}_{k_i}, \mathbf{e}_{k_j} \rangle / \sigma_e^2 \in [0, 1)$ *(or a common non-zero mean* $\boldsymbol{\mu}$*), the exact variance-preserving factor generalizes to*

$$\alpha_{gen}(\rho) = \sqrt{\frac{(1-\rho)\sum_{i=1}^{K} w_i^2 + \rho\left(\sum_{i=1}^{K} w_i\right)^2}{(1-\rho)\sum_{i=1}^{k} w_i^2 + \rho\left(\sum_{i=1}^{k} w_i\right)^2}},$$

*which smoothly interpolates between* $\alpha_{L_2}$ *(*$\rho \to 0$*) and* $\alpha_{L_1}$ *(the degenerate collinear limit* $\rho \to 1$*). Moreover, although* $\alpha_{proj} = 1$ *minimizes pre-normalization Euclidean MSE, the FFN output is immediately normalized by RMSNorm*$(\mathbf{r} + \mathbf{y}_{FFN})$*. Leaving* $\alpha = 1$ *reduces the variance of the routed MoE branch relative to the residual stream* $\mathbf{r}$ *and shared expert* $\mathbf{e}_{shared}$*, whereas* $\alpha_{L_2}$ *preserves the exact signal-to-noise energy ratio between the routed and shared branches prior to* `RMSNorm` *division.*

Table 3 validates Proposition 1 empirically on `DeepSeek-V4-Flash 284B` on a deterministic code-generation micro-benchmark (24 teacher-forced single-token decode steps over an algorithmic Python/C prompt plus 180-token free generation, where $D_{\mathrm{KL}}(P\|Q)$ and the local perplexity shift $\Delta\mathrm{PPL}_{\text{top-10}} = \exp(D_{\mathrm{KL}}) - 1$ are evaluated over the renormalized top-10 logit support): switching from $\alpha_{L_1}$ to $\alpha_{L_2}$ at $k = 4$ **cuts top-10 next-token KL-divergence by 58.1%** ($0.0671 \to 0.0281$ nats) and restores 100.0% Top-3 token agreement, while $k = 5$ ($\alpha_{L_2}$) achieves **9.68 tok/s** with a negligible **0.0042 nats top-10 KL-divergence (**$+0.42\%$ **local** $\Delta\mathbf{PPL}_{\text{top-10}}$**)** and **0.99714 probability cosine similarity**.

### 3.3 L1-Local Stratified Gate-First SwiGLU Row Skipping with $L_2$ Gate-Energy Preservation

In every SwiGLU FFN block (both the 43 shared experts `ffn_*_shexp` and the $43 \times K$ active routed experts `ffn_*_exps`), the intermediate state is $\mathbf{h} = \mathrm{SiLU}(\mathbf{W}_{\text{gate}}\mathbf{x}) \odot (\mathbf{W}_{\text{up}}\mathbf{x})$ with $h_j = \frac{g_j}{1+e^{-g_j}} u_j$ ($j \in \{0, \ldots, 2047\}$), followed by $\mathbf{y} = \mathbf{W}_{\text{down}}\mathbf{h}$. Because ggml_swiglu_split(gate, up) binds gate to `src[0]` and up to `src[1]`, `ggml`'s post-order DFS schedules $\mathbf{g} = \mathbf{W}_{\text{gate}}\mathbf{x}$ immediately before $\mathbf{u} = \mathbf{W}_{\text{up}}\mathbf{x}$, and the live tensor `glu_node->src[0]` is guaranteed by graph topology to hold $\mathbf{g}$ throughout the execution of $\mathbf{W}_{\text{up}}\mathbf{x}$.

Furthermore, because $d_{ff} = 2048$ is partitioned identically across $T = 16$ pinned physical cores into 128-row strata $\mathcal{C}_c = [128c, 128c + 127)$, thread $c$'s slice of $\mathbf{g}$ resides in Core $c$'s private 32 KB L1D cache when $\mathbf{W}_{\text{up}}\mathbf{x}$ begins. Since $\max_{g \le 0} |\mathrm{SiLU}(g)| = 0.2785$, near-zero and negative gates suppress $u_j$ regardless of its magnitude. In `ggml_swiglu_gate_mask_128`, each thread $c$ executes an $O(128)$ in-register Hoare quickselect on $s_j^2 = \mathrm{SiLU}(g_j)^2$ ($< 40$ ns), skips fetching the bottom $128 - m$ contiguous rows of $\mathbf{W}_{\text{up}}$ from DDR5 DRAM (800 B/row in `IQ1_S`, 2,816 B/row in `Q5_K`), and rescales the $m$ kept rows by the conditional $L_2$ gate-energy factor:

$$\alpha_{\text{SwiGLU}}^{(c)} = \sqrt{\frac{\sum_{j \in \mathcal{C}_c} \mathrm{SiLU}(g_j)^2}{\sum_{j \in \mathcal{A}_c} \mathrm{SiLU}(g_j)^2}}, \quad |\mathcal{A}_c| = m. \tag{7}$$

Conditioned on $\mathbf{g}$, $\mathbb{E}[h_j^2 \mid g_j] = \mathrm{SiLU}(g_j)^2 \sigma_u^2$, so rescaling by $\alpha_{\text{SwiGLU}}^{(c)}$ preserves 99.86%–100.00% of the exact output norm $\|\mathbf{W}_{\text{down}}\mathbf{h}\|_2$ while ensuring zero straggler threads at the threadpool barrier and zero cross-thread synchronization. Controlled dynamically via `cfg[6]` (`exps`) and `cfg[7]` (`shexp`) in `/tmp/dsv4_moe_cfg.bin` for $l \ge 3$, combining `Top-5/Top-4` ($\alpha_{L_2}$) with Gate-First SwiGLU reaches **10.15–10.74 tok/s** (93.15 ms/tok) with 100% exact agentic JSON tool-call agreement (Table 3, rows 7–9).

## 4 Draftless Speculative Decoding: `RouteCrack + Stockfish-PVS v2`

### 4.1 The CPU-MoE Speculative Golden Equation

In standard speculative decoding [4], $D$ draft tokens are verified in a single batched target forward pass. On a GPU where $D \le 8$ lies inside the compute-bound roof, verification time equals a single-token step ($c_{\text{gpu}} \approx 0$). On a memory-bandwidth-bound CPU hosting a Sparse MoE, each additional token in a verification batch activates additional routed experts from DDR5 ($\sim 0.30\times$ single-token latency per drafted token on `Nemotron-3.5-30B-A3B`), and any rejected draft token triggers an 18.4 MB Mamba-2 recurrent state rollback (`slot.spec_ckpt`). Consequently, net wall-clock gain over $A$ accepted tokens and $W = D - A$ wasted tokens obeys:

$$\mathrm{WallGain}_{\text{CPU}} = A - 0.30\,D = 0.70\,A - 0.30\,W. \tag{8}$$

A 1B neural draft model reading 600 MB per step incurs $\sim 40$ ms of DDR5 latency for $D = 5$ and flushes the 64 MB L3 cache, yielding 11.5 tok/s (slower than non-speculative decoding).

### 4.2 `StockfishDrafter`: 192 KB L2 Zobrist TT & Principal Variation Search

To achieve high recall without DRAM traffic, `RouteCrack` (`rc_predictor.h`) pairs a static 8-level FNV-64 hash cascade, an online session trie (`rc_trie_v2.h`), and an 11,382-entry multi-token macro codebook with **`StockfishDrafter` (`rc_stockfish.h`)** [5]. `StockfishDrafter` maintains three lockless 4-way bucketed Zobrist transposition tables (`tt5`, `tt4`, `tt3` $= 3 \times 64$ KB $= 192$ KB, $< 19\%$ of Core 0's 1 MB L2 cache on Zen 5) with 32-bit verification keys, verified `best_move`, explicit `refuted_move` tracking, and history gravity ($\in [-512, +512]$).

Table 3: Empirical quality vs. decode speed of runtime MoE tail pruning on `DeepSeek-V4-Flash 284B` ($l \in \{0..2\}$ kept at 6/6 experts; evaluated over 24 teacher-forced code-generation decode steps + 180-token free generation on 16C Zen 5, with $D_{\mathrm{KL}}$ and $\Delta\mathrm{PPL}_{\text{top-10}} = \exp(D_{\mathrm{KL}}) - 1$ measured on the renormalized top-10 logit support).

| Runtime MoE Configuration | Decode (tok/s) | Latency | Speedup vs. Init | Prob CosSim | Top-10 $D_{\mathrm{KL}}$ (nats) | Local $\Delta\mathrm{PPL}_{\text{top-10}}$ | Top-1 / Top-3 Match |
|---|---|---|---|---|---|---|---|
| 0. `v2.3.0` Initial Baseline ($N = 1$) | 5.91 tok/s | 169.30 ms | 1.00× | 1.00000 | 0.0000 | +0.00% | 100.0%/100.0% |
| 1. **`v2.4.0` Lossless (`Top-6` All, Default)** | **9.06 tok/s** | **110.33 ms** | **+53.3%** | **1.00000** | **0.0000** | **+0.00%** | **100.0%/100.0%** |
| 2. **Adaptive Tail** ($w_i < 0.11$, $\alpha_{L_2}$) | **9.52 tok/s** | **105.02 ms** | **+61.1%** | **0.99634** | **0.0047** | **+0.47%** | **95.8%/100.0%** |
| 3. **Sweet-Spot (`Top-5` on $l \geq 3$,** $\alpha_{L_2}$) | **9.68 tok/s** | **103.27 ms** | **+63.8%** | **0.99714** | **0.0042** | **+0.42%** | **95.8%/100.0%** |
| 4. **Hybrid `6/4/6`** ($l \in [16, 39]$ **at 4,** $\alpha_{L_2}$) | **9.83 tok/s** | **101.74 ms** | **+66.3%** | **0.99660** | **0.0184** | **+1.86%** | **95.8%/100.0%** |
| 5. **Hybrid `5/4/5`** ($l \in [12, 40]$ **at 4,** $\alpha_{L_2}$) | **9.95 tok/s** | **100.55 ms** | **+68.4%** | **0.99463** | **0.0215** | **+2.17%** | **95.8%/100.0%** |
| 6. **Turbo `10+` (`Top-4` on** $l \geq 3$, $\alpha_{L_2}$) | **10.12 tok/s** | **98.84 ms** | **+71.2%** | **0.99587** | **0.0281** | **+2.85%** | **95.8%/100.0%** |
| *Ablation: `Top-4` with Linear $L_1$ Rescaling ($\alpha_{L_1}$)* | 10.10 tok/s | 98.97 ms | +70.9% | 0.98698 | 0.0671 | +6.94% | 95.8%/95.8% |
| 7. `Top-5` ($\alpha_{L_2}$) + **Gate-First SwiGLU** (80/128) | **10.15 tok/s** | **98.49 ms** | **+71.7%** | **0.99385** | **0.0531** | **+5.45%** | **100.0%/100.0%** |
| 8. `Top-4` ($\alpha_{L_2}$) + **Gate-First SwiGLU** (96/128) | **10.60 tok/s** | **94.37 ms** | **+79.4%** | **0.99126** | **0.0694** | **+7.19%** | **100.0%/100.0%** |
| 9. `Top-4` ($\alpha_{L_2}$) + **Gate-First SwiGLU** (64/128) | **10.74 tok/s** | **93.15 ms** | **+81.7%** | **0.98059** | **0.1159** | **+12.29%** | **100.0%/100.0%** |

Crucially, because static cascade tables and live-session Zobrist transposition tables have *uncorrelated error modes*, gating uncertain candidates on their orthogonal agreement unlocks four previously unusable high-recall predictors: (1) Online Session Trie Rescue (92.19% marginal precision), (2) Top-4 Cascade Disambiguation, (3) Multi-Token Macro Codebook chains (96.44% precision), and (4) **Singular Horizon Extension up to $M = 7$ tokens/step** (98.74% marginal precision at $D = 5..7$). As shown in Table 4 (evaluated across 77,456 held-out tokens in `test_synergy.cpp`), `Stockfish-PVS v2` adds +223 **accepted tokens with zero net additional wasted tokens ($W = 892 \rightarrow 892$, 100.0% net marginal precision)** and eliminates 32 Mamba-2 rollbacks.

Table 4: Held-out ablation across 77,456 `Nemotron-3.5` tokens (351 contexts, `speculative/cpp/test_synergy.cpp`).

| Drafter Configuration | Draft $D$ | Acc $A$ | Waste $W$ | Rollbacks | WallGain |
|---|---|---|---|---|---|
| 1. `RouteCrack` ($M = 5$) | 6,332 | 5,440 | 892 | 482 | +3,540.4 |
| 2. `+ Stockfish-PVS v1` | 6,329 | 5,459 | **870** | 461 | +3,560.3 |
| 3. **`+ Stockfish-PVS v2`** ($M = 7$) | **6,555** | **5,663** | **892** | **450** | **+3,696.5** |

# 5 Microarchitectural & Multi-Platform Systems Co-Design

## 5.1 Kernel & Graph Optimizations in `ggml-cpu`

1. **Zero-Rebuild Dynamic `CSA/LID` Skip (`ggml_dsv4_skip_csa_comp`)**: In `DeepSeek-V4-Flash`, `CSA/LID` compressor blocks execute only once every 8 tokens during decode. Instead of rebuilding the 4,551-node compute graph when transitioning between compressor and non-compressor steps, we build the superset graph once, tag dummy compressor nodes with `flags |= 0x100`, and skip them in $O(1)$ inside `ggml_graph_compute_thread`, eliminating 26.8 ms/tok of graph allocation overhead.

2. **8-Slot Thread-Local `Q8_K`/`Q8_0` Activation Cache & `MUL_MAT_ID` Row-Group Reuse**: When multiple projections share the same FP32 input tensor (e.g., `wq_a` + `wkv_a`, or `up_exps` + `gate_exps` + `shexp`), `s_q8_cache_buf` quantizes the activation once and reuses it across subsequent contractions, while `g_tls_mmid_ids_valid` reuses the bucketed expert row mapping between `up_exps` and `gate_exps`.

3. **In-Register 9-Stage AVX2 Fast Walsh-Hadamard Transform & Vectorized `HC_PRE/POST`**: `DeepSeek-V4-Flash` invokes a 512-point FWHT (`GGML_OP_HADAMARD`) 396 times per token. We keep all 9 butterfly stages ($s = 1, 2, \ldots, 256$) inside AVX2 `YMM` registers using lane permutes (`_mm256_permute2f128_ps`, `_mm256_shuffle_ps`) and vectorize the 4-stream Sinkhorn-20 Hyper-Connections (`HC_PRE/POST`), cutting non-matrix operator time in half (19.60 → 9.85 ms/tok).

4. **2 MB Transparent HugePages, 3D L1D Tiling, & Bidirectional `_MM_HINT_NTA` L3 Isolation**: Allocating weight maps and compute arenas with 2 MB-aligned `MADV_HUGEPAGE` expands Zen 5 L2 STLB coverage to 6.14 GB (eliminating 99.6% of page-table walks). Streaming cold MoE weights and KV tokens older than `LLAMA_KV_L3_WINDOW` (2,048 on Nemotron; 1,024 on DeepSeek MLA with 8-step sparse lookahead prefetching) via `_mm_prefetch(..., _MM_HINT_NTA)` prevents L3 victim-cache pollution.

5. **4-Level Register (`ZMM` AVX-512BW) + L1D (32 KB) + L2 (1 MB) + L3 (64 MB) Multi-Column Prefill Tiling & `MQA8` FlashAttention**: To overcome the DDR5 bandwidth wall during prefill ($N > 1$) and multi-turn agentic workflows, we implement 512-bit AVX-512BW Register-Stationary multi-column kernels (`_1x8`, `_1x4`, `_1x3`, `_1x2`) across all six quantized formats (`Q2_0`, `IQ1_S`, `IQ2_XXS` with 7-instruction BMI2+SWAR parity masks, `IQ1_M`, `Q5_K`, `Q6_K`), an AVX-512 `quantize_row_q8_K` vectorizer, and a Register-Stationary AVX-512 `MQA8` FlashAttention kernel (2.0× faster). Decomposing any column count $M \geq 2$ into $\{8, 4, 3, 2\}$ eliminates

$1 \times 1$ fallback across ragged 256-expert MoE buckets, accelerating `DeepSeek-V4-Flash 284B` prefill from 25.05 tok/s to **43.16–44.36 tok/s (+77.1%, 100% bit-exact)** while maintaining **9.16–9.95 tok/s** lossless decode across multi-turn agentic sessions.

6. **Cross-Platform Thread-Safe Runtime Control (`Linux/macOS mmap` & `Windows _WIN32`)**: Runtime MoE parameters ($k_{\text{early}}, k_{\text{mid}}, w_{\text{min}}$, scale_mode) are initialized on thread 0 with release-acquire atomic synchronization (`atomic_int s_moe_topk`, `s_shared_moe_skip_mask`) and read from a 32-byte shared file mapping (`/tmp/dsv4_moe_cfg.bin` with `O_NOFOLLOW` and `DSV4_MOE_CFG_PATH` override via POSIX `mmap`, and `%TEMP%\dsv4_moe_cfg.bin` via Win32 `CreateFileMappingA`/`MapViewOfFile`), enabling sub-nanosecond (0.3 ns) L1-cached inspection with zero syscalls per token while resetting the skip mask across batched prefill passes ($N > 1$).

### 5.2 Multi-Chiplet Scaling & Hybrid GPU+CPU Tensor-Role Splitting

Table 5 summarizes cross-platform scaling:

- **8-CCD Threadripper PRO 7985WX (4C/CCD Law)**: Launching 64 threads (8C/CCD) across 8 CCDs saturates the Infinity Fabric with atomic work-stealing traffic, dropping decode speed from 80.14 tok/s to 51.80 tok/s. Selecting 4 physical cores per CCD (`-t 32`) via sysfs L3 topology discovery yields **386.00 tok/s prefill**, **80.14 tok/s cold decode**, and **111.62 tok/s warm decode**.
- **Hybrid GPU+CPU Tensor-Role Splitting (`-ngl 99 -n-cpu-moe` $N$ `-load-mode none`)**: On 11–16 GB GPUs (Linux Vulkan or Windows CUDA), offloading by tensor role keeps 100% of the dense backbone, KV cache, and Mamba-2 `SSM_SCAN` in high-bandwidth VRAM while placing $N$ sparse MoE layers in System RAM. On an AMD RX 9070 XT 16GB, `Fractal-MXFP4` (15.2 GiB) achieves **75.1 tok/s (new code) / 80.8 tok/s (in-context refactor)** at 256K context, outperforming `Q4_0` (69.7 / 76.2 tok/s at 16.1 GiB) while saving 0.9 GB of VRAM. On Windows 10/11 (`windows/modchip_win.ps1`), live free VRAM is queried via `nvidia-smi` with a 550 MB WDDM safety reserve to prevent PCIe GTT paging.

## 6 Limitations & Negative Results

1. **CPU-Only Scope of `Fractal-64` (`Q2_0`)**: Because `Fractal-64` repurposes the `Q2_0` type ID with a custom 30-byte Block-64 layout supported in `ggml-cpu`, offloading `Q2_0` MoE tensors to Vulkan/CUDA GPUs without a matching GPU shader produces invalid outputs; hybrid GPU+CPU setups should use `Fractal-MXFP4` or `Q4_0`, which have native Vulkan and CUDA shaders.
2. **Non-Temporal Prefetching on 50-Byte Unaligned `iq1_s` Blocks**: Issuing `_MM_HINT_NTA` inside `ggml_vec_dot_iq1_s_q8_K` degraded `iq1_s` throughput by $\sim 5\%$ because `sizeof(block_iq1_s)` $= 50$ bytes ($< 64$-byte cache line), causing premature eviction of cache lines straddling adjacent blocks.
3. **Catastrophic Cliff Below Top-4 Routing**: While $L_2$-quadrature pruning from Top-6 to Top-5 (+0.42% PPL) or Top-4 (+2.85% PPL) preserves generation quality, truncating to Top-2 alters the covariance spectrum across 43–52 layers and causes EOS suppression.

## 7 Conclusion

By co-designing hierarchical non-linear quantization (`Fractal-64` and analytical `SwiGLU-H2`), AVX-512/AVX2 kernels, L2/L3-resident chess-engine speculative search (`Stockfish-PVS v2`), and $L_2$-quadrature MoE / low-rank mathematics, `llama-modchip` enables frontier 30B–284B Sparse MoE inference at 256K context on commodity 64 GB CPUs and consumer GPU+CPU hybrids. Full source code, C/C++ patches, and reproducible benchmarks are available at https://github.com/synchopate/llama-modchip.

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Table 5: Multi-platform verification across x86-64 (Zen 5, 8-CCD Zen 4 Threadripper), ARM64 (AWS Graviton3, Apple M4 Pro), and GPU+CPU Hybrids (Linux/Windows Vulkan & CUDA) on `Nemotron-3.5-30B-A3B` and `DeepSeek-V4-Flash 284B`.

| Hardware Platform | Cores / Cache / GPU | Memory | Model & Profile | Prefill (tok/s) | Base / Cold Decode | Warm / Burst Decode |
|---|---|---|---|---|---|---|
| AMD Threadripper PRO 7985WX | 64C/128T (8 CCDs, 256MB L3) | 256GB 8-Ch DDR5 | Nemotron-30B (`-t 32` 4C/CCD) | **386.00** | **73.84 – 80.14** | **111.62 tok/s** |
| AMD 7945HX3D + RX 9070 XT | 16C (128MB L3) + 16GB Vulkan | 94GB DDR5 + 16GB | Nemotron-30B (`MXFP4` 256K) | **563.00** | **75.10 tok/s** | **80.80 tok/s** |
| AMD Ryzen 9 9955HX (Zen 5) | 16C/32T (2 CCDs, 64MB L3) | 64GB DDR5 (0 GPU) | Nemotron-30B (`Fractal-64`) | **241.0 – 306.8** | **25.02 – 34.52** | **48.39 / 60.40 tok/s** |
| Apple M4 Pro (Unified SoC) | 14C CPU + 20C Metal GPU | 24GB Unified (0 Swap) | Nemotron-30B (128K Ctx) | **256.0 – 549.0** | **47.0 – 48.0** | **49.0+ tok/s** |
| AWS Graviton3 `m7g.2xlarge` | 8 vCPU ARM64 Neoverse-V1 | 31GB DDR5 (0 GPU) | Nemotron-30B (`-t 8`) | **101.67** | **41.05 tok/s** | **44.97 tok/s** |
| Intel i9-9900K + RTX 2080 Ti | 8C/16T + 11GB CUDA (Win 11) | 16GB DDR4 + 11GB | Qwen3-14B / Nemotron-8B / 30B | Auto CUDA 12.4 | **50.0 – 58.0** | **100+ tok/s** (8B) |
| AMD Ryzen 9 9955HX (Zen 5) | 16C/32T (2 CCDs, 64MB L3) | 64GB DDR5 (58.9GiB RSS) | **DeepSeek-V4 284B** (`Fractal`) | **43.16 – 44.36** (32.6ms Slot0) | **9.06 – 10.74** | **11.50 – 14.69 tok/s** |


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